# Replication of *The Efficacy of Seven Gratitude Interventions for Promoting Subjective Well-Being* by Dang et al. (2025, *Journal of Positive Psychology*)

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### Introduction

Dang et al. (2025) sought to test 7 gratitude interventions that are used in positive psychology to improve subjective well-being. These interventions included intrapersonal, interpersonal, and gratitude-to-God interventions. While studies in the literature have examined these interventions before, such studies are heterogeneous in their selection of outcome measures for subjective well-being and only compare a few interventions against each other, making comparisons across excluded interventions difficult. By contrast, this study included a comprehensive battery of interventions and standardized outcome measures (i.e., state gratitude, state positive/negative

affect, and life satisfaction). As a result, this experiment allows for a direct comparison between the efficacy of gratitude interventions and can inform what gratitude practices are most helpful for the general public to incorporate into their lives.

In Study 1, Dang et al. (2025) found that writing a private gratitude letter was the most effective intervention in increasing positive affect. In this replication report, I wanted to replicate this finding and hoped to find a similarly large Hedges' $g$ of around 0.41 when comparing this intervention to an active control. An important difference between this replication and the original study is that this effect was tested in a student population rather than US adults from Prolific. This was aimed to increase the generalizability of the original study and offer guidance for students to improve their well-being.

## Methods

### Power Analysis

The key effect size for this replication (gratitude letter vs. events list) was a Hedges' $g = 0.41$. Hedges' $g$ can be thought of as a bias-corrected form of Cohen's $d$ which is used because Cohen's $d$ is upwardly biased (especially in small samples). While the Dang et al. (2025) study reported the unbiased Hedges' $g$, many power analysis programs only offer sample size estimates based on Cohen's $d$. With larger samples, the difference between Hedges' $g$ and Cohen's $d$ becomes minimal as the $J(x)$ correction factor applied to Cohen's $d$ approaches 1 with large $n$ . However, for the sake of precision, the Hedges' $g$ was converted to a Cohen's $d$ for the power/sample size analysis. The conversion is explained below:

Using the Hedges' $g$ to Cohen's $d$ conversion, we can divide Hedges' $g$ by the $J(x)$ correction factor to recover Cohen's $d$, where $x = n_1 + n_2 - 2$ for the correction factor $J(x)$. In Study 1, both conditions had exactly $n_1 = n_2 = 126$ participants, therefore $x = 126 + 126 - 2 = 250$.

$$
\begin{aligned}
J(x) \cdot d &= g \\
d &= \frac{g}{J(x)} \\
d &= \frac{g}{\frac{\Gamma(x/2)}{\sqrt{x/2} \times \Gamma((x-1)/2)}}
\end{aligned}
$$

$$
\begin{aligned}
&\#\text{Substitute } J(x) = J(250) \\
d &= g \times \frac{\sqrt{250/2} \times \Gamma((250-1)/2)}{\Gamma(250/2)} \\
d &= g \times \frac{\sqrt{125} \times \Gamma(124.5)}{\Gamma(125)} \\
d &= g \times 1.003013...
\end{aligned}
$$

```
# Calculate Correction Factor
J_250 <- sqrt(125) * gamma(124.5) / gamma(125)
J_250
```

```
[1] 1.003013
```

```
ComputedCohens_d = 0.41 * J_250
ComputedCohens_d
```

```
[1] 0.4112351
```

Now that the Cohen's $d$ has been computed ($d = 0.4112351$), we used the `pwr` package in R to perform a sample size analysis for 80%, 90%, and 95% power.

```
# Step 1: Set CRAN as the source to install `pwr`
options(repos = c(CRAN = "https://cloud.r-project.org"))

# Step 2: Install `pwr`
install.packages("pwr")
```

```
The downloaded binary packages are in
    /var/folders/0p/p3phj575333gmy1dhsm2fsqr0000gp/T//RtmpRw0MIE/downloaded_packages
```

```r
# Step 3: Load `pwr` into the current session
library(pwr)
```

### 80% Power

```r
Power80 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .80,
           type = "two.sample",
           alternative = "two.sided")
```

```r
Power80$n
```

```
[1] 93.79321
```

At an alpha level of .05, for 80% power, we are going to need 93.7932103 participants per group. In total, that will be 188 participants.

### 90% Power

```r
Power90 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .90,
           type = "two.sample",
           alternative = "two.sided")
```

```r
Power90$n
```

```
[1] 125.2323
```

At an alpha level of .05, for 90% power, we are going to need 125.2322846 participants per group. In total, that will be 251 participants.

### 95% Power

```
Power95 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .95,
           type = "two.sample",
           alternative = "two.sided")
```

At an alpha level of .05, for 95% power, we are going to need 154.6466419 participants per group. In total, that will be 310 participants.

### Planned Sample

For a balance of power and practicality, we will aim for a minimum power level of 80% which will require 188 participants. (# This should be feasible in a sample of SONA students especially since mine is an online survey experiment that is less than 30 minutes.)

The participant pool will be undergraduates from the University of California San Diego. There likely will be no adverse effects of the study. However, if any are reported by participants, data collection will be stopped.

While we will not exclude human participants, we do not want to include bots or responses from LLMs and as a result, invisible white text will be added into the prompt in an attempt to catch AI responses. It will ask the participant to compute the result of gamma(125) and discretely put that result in the response without decimal points (i.e., prompt injection). Responses that include the $1.5061417 \times 10^{207}$ will be row-wise deleted.

### Materials

Study materials include the jsPsych code used to run the experiment. Participants will also need access to a computer and an internet connection.

### Procedure

In the interest of replication, I will use the same instructions that were provided in the original study (Cunha et al., 2019; Dang et al., 2025; Emmons & McCullough, 2003; Walsh et al., 2022). Participants will be randomly assigned to two conditions and see the following directions:

- **Control Group: Events List (Cunha et al. 2019):** "In life, there are events, both large and small, that end up affecting us. Think back over the past day and write down five events that somehow affected you."

- **Experimental Condition: Gratitude Letter (Walsh et al., 2022):** "In our daily lives, we often thank people for both large and small things. However, it can sometimes be difficult to make time to express our gratitude more deeply. Today, we want you to write a letter of gratitude to someone who has done something for which you are extremely grateful. Please do not share your letter with this person or anyone else."

Participants will also be told: > As you write, don't worry about perfect grammar and spelling, and remember that anything you write will remain strictly confidential. Should an experimenter read this entry in the future, it will be identifiable only by a participant number and not by your name'. (Dang et al., 2025).

After they finish the intervention, they will be asked to fill out a questionnaire that asks them about their state positive affect, some demographic questions, and will be thanked for their participation in this study. Participants will be awarded research credits that can be applied to fulfill course research participation requirements or confer extra credit.

## Measures

### State Positive Affect

Positive affect will be assessed the same way as in the original study. Using a 7 point likert-scale from (1) *strongly disagree* to (7) *strongly agree*, participants rated how much they agree with five adjectives: "happy," "joyful," "pleased," "content," and "satisfied" (Dang et al., 2025). The original study demonstrated really high internal consistency for this measures, McDonald's Omega = .95.[1]

### State Gratitude

Using a 7 point likert-scale from (1) *strongly disagree* to (7) *strongly agree*, participants rated how much they agree with three adjectives: "grateful," "appreciative," and "thankful" (Dang et al., 2025). This scale likewise demonstrated good internal consistency in the original study, McDonald's Omega = .93.

### Demographics

Participants were asked questions about their age, gender, race, and ethnicity.

[1] McDonald's Omega describes the proportion of variance in the indicators that is explained by a general latent factor compared to the overall amount of variance in the indicators with the highest score being 1.

$$\omega = \frac{\left(\sum_{i=1}^{k} \lambda_i\right)^2}{\left(\sum_{i=1}^{k} \lambda_i\right)^2 + \left(\sum_{i=1}^{k} \varepsilon_i\right)}$$

### Analysis Plan

#### Statistical Software and Analyses

For the data analysis, I will use the R programming language (R Core Team, 2026). Aligned with the power analysis for sample size, I will set the alpha threshold to $\alpha = .05$ and use the `t.test` function to compute the $t$-test test statistic. This function will also output a Cohen's $d$. To calculate the Hedges' $g$ effect size, I will apply the Hedges correction $J(x)$ depending on the degrees of freedom that actually end up in the final analyses. Interestingly, the original study did not correct for multiple testing despite the enormous amount of pair-wise comparisons ($\alpha = .05$ was used across analyses). I will not need to worry about multiple testing in this case as I am only making one comparison/test.

The original study used an intent-to-treat analysis which meant that they did not exclude any participants from the study. This is aimed to measure the generalizable real-world impact of an intervention and acknowledge that attrition and non-compliance are inseparable downstream effects of the intervention. Since this study is meant to inform real student populations, it is vital to understand the realistic intent-to-treat effect size as opposed to the effect size found in an artificially controlled setting. Thus, I opted to also use an intent-to-treat analysis technique.

#### Differences from Original Study

Key differences from the original study mainly include using SONA for data collection among UC San Diego undergraduates and only having 2 conditions as opposed to the 11 in the original study.

## Results

### Data Preparation

Aligned with the Dang et al. (2025) paper analysis protocol, data will be analyzed without excluding participants to measure the intent-to-treat effect. Prior to running analyses, responses will be scanned for AI flags (viz., the gamma(125) AI catch). Participants that fail the AI check will be excluded entirely and be row-wise deleted. Individual responses will also be checked prior to dropping to ensure that responses did not actually contain that number for legitimate reasons.

```
library(dplyr)
Tensor <- read_csv("")

# Look at the rows that have these in there to drop
```

```
Tensor <- Tensor[grepl("1.5", Tensor$COL), ]
Tensor <- Tensor[grepl("15", Tensor$COL), ]

# Drop the appropriate participant right here.
```

```
gamma(125)
```

```
[1] 1.506142e+207
```

## Confirmatory Analysis

```
MainEffect <- t.test(DV ~ Condition, data = Tensor)

# Calulate Hedges' g using J(x) and Cohen's d
```

A $t$-test was computed to assess whether the average positive affect score differed between the gratitude letter and events list conditions, $t(df) = XX.XX$, $p < or = .XX$, Hedges' $g = GG.GG$. There was/was not a significant difference such that the _____ group had higher mean positive affect scores than the _____ group.

## Exploratory Analyses

Using latent factor scores for positive affect and gratitude, construct a mediation model that tests whether gratitude may be a putative mediator.

```
Syntax for SEM

PA BY PA1 PA2 PA3 PA4 PA5
GRAT BY GRAT1 GRAT2 GRAT3

PA ON GRAT COND
GRAT ON CONDf
```

“‘

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